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Question 1. Relatin' Canes

(a)

Define $R \subset \mathbb{Z} \times \mathbb{Z}$ by $(a, b) \in R$ iff $a + 2b$ is odd.

We know a must be odd, since $2b$ is even. Therefore, R is:

- **Reflexive:** $(2, 2) \notin R$
- **Symmetric:** $(1, 2) \in R$ but $(2, 1) \notin R$
- **Antisymmetric:** $(1, 3) \in R$ and $(3, 1) \in R$
- **Transitive**

(b)

Define $S \subset \mathbb{N} \times \mathbb{N}$ by $(a, b) \in S$ iff $a \cdot b \geq 0$.

S is:

- **Reflexive**
- **Symmetric**
- **Antisymmetric:** $(1, 2) \in S$ and $(2, 1) \in S$
- **Transitive**

(c)

Let $A = \{k \in \mathbb{Z} : k < 0\}$ be the set of negative integers. Define $T \subset A \times A$ by $(a, b) \in T$ iff $a \times b < 0$.

T is:

- **Reflexive:** $(-1, -1) \notin T$
- **Symmetric**
- **Antisymmetric**
- **Transitive**

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(d)

let $B = \mathcal{P}(\mathbb{Z})$. Define $U \subset B \times B$ by $(X, Y) \in U$ iff $X \cup \mathbb{N} \subset Y \cap \mathbb{N}$.

For $X \cup \mathbb{N} \subset Y \cap \mathbb{N}$ to be true:

- $\mathbb{N} \subset Y$
- $X \subset \mathbb{N}$

Therefore, U is:

- **Reflexive:** Suppose $(X, Y) = (\{1\}, \{1\})$. Then $X \cup \mathbb{N} = \mathbb{N}$ and $Y \cap \mathbb{N} = \{1\}$. Since $\mathbb{N} \not\subset \{1\}$, $(\{1\}, \{1\}) \notin U$.
- **Symmetric:** $(\{1\}, \mathbb{N}) \in U$ but $(\mathbb{N}, \{1\}) \notin U$
- **Antisymmetric**
- **Transitive**

Question 2. Better Get Proving

(a)

Let a and b be arbitrary elements.

Suppose $(a, b) \in R \circ R$ and $(b, a) \in R \circ R$.

By the definition of compose, there is some element c such that $(a, c) \in R$ and $(c, b) \in R$.

Also for the latter, by the definition of compose, there is some element d such that $(b, d) \in R$ and $(d, a) \in R$.

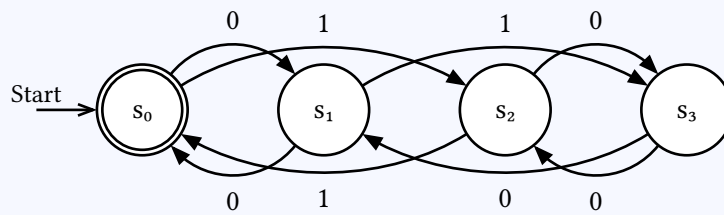
We're given that R is transitive. Since $(a, c) \in R$ and $(c, b) \in R$, then by the definition of transitivity, $(a, b) \in R$. Furthermore, since $(b, d) \in R$ and $(d, a) \in R$, then by the definition of transitivity, $(b, a) \in R$.

We're also given that R is antisymmetric. Since $(a, b) \in R$ and $(b, a) \in R$, then by the definition of antisymmetric, $a = b$.

Therefore, since a and b were arbitrary, we have proven, by the definition of antisymmetric, that $R \circ R$ is antisymmetric.

Question 3. Don't Stop BelEven'

(a)



- s_0 : Even number of 0's and even number of 1's
- s_1 : Odd number of 0's and even number of 1's
- s_2 : Even number of 0's and odd number of 1's
- s_3 : Odd number of 0's and odd number of 1's

(b)

$$(00 \cup 11 \cup ((01 \cup 10)(00 \cup 11)^*(01 \cup 10)))^*$$

(c)

S is a CFG that generates M :

$$S \rightarrow SS \mid 0T1 \mid 1T0 \mid \varepsilon$$

$$T \rightarrow ST \mid TS \mid 0S1 \mid 1S0$$

- S generates strings with the same number of 0's and 1's and an even number of 0's and 1's.
- T generates strings with the same number of 0's and 1's and an odd number of 0's and 1's.